FinPal - 20-Day Detailed Build Plan

PayPal AI Hackathon | 15 October - 3 November core build | Submission: 12 November 2026
Team: Dave + Henry (Frontend) - Celestine & Steph (Backend)

Locked stack: Next.js App Router monorepo, Supabase Auth/Postgres, PayPal Sandbox central merchant credentials, Channel3, AG Grid, and Render for backend/API. Bryntum and KERNEL are P1. No FastAPI and no user PayPal OAuth.

Delivery rule

The core loop is proposal -> constraints -> Quality Score -> decision -> authorization/capture -> audit. Do not begin extra sponsor integrations until that loop is stable.

Day 1 - Foundation (Thu 15 Oct)

- Create the Next.js monorepo, Tailwind/shadcn shell, shared domain types, linting, tests, README, and .env.example.
- Create Supabase project, Auth flow, initial migrations, row-level-security plan, and Render service.
- Provision central PayPal Sandbox merchant credentials and Channel3 access. Agree on API contracts and daily sync.

Day 2 - PayPal State Model (Fri 16 Oct)

- Implement CREATED -> APPROVED -> AUTHORIZED using idempotent route handlers.
- Persist PayPal IDs and exact uppercase states. Add void from AUTHORIZED; never call a post-capture reversal a void.
- Build settings/navigation skeleton and sandbox disclosure.

Day 3 - Constraints and Schema (Sat 17 Oct)

- Create users, constraints, proposals, evaluations, decisions, payment_attempts, and append-only audit_events migrations.
- Implement time-window and amount-cap checks with unit tests.
- Build the constraint settings form.

Day 4 - Audit Foundation (Sun 18 Oct)

- Create the insert-only audit service and entity event taxonomy.
- Enforce RLS and expose a user-scoped decisions/activity endpoint.
- Build an initial activity list before AG Grid integration.

Day 5 - Channel3 Data (Mon 19 Oct)

- Integrate product search and normalized comparable offers.
- Handle currency mismatch, fewer than three offers, timeouts, and provider errors.
- Display raw reference offers on a test page.

Day 6 - Quality Score v1 (Tue 20 Oct)

- Implement deterministic thresholds: Good <= median x 1.05; Fair <= median x 1.20; otherwise Poor.
- Return Insufficient data for fewer than three usable offers and prevent auto-approval.
- Persist median, lowest offer, offer count, currency, references, and explanation. Add boundary tests.

Day 7 - Decision Orchestrator (Wed 21 Oct)

- Implement the locked decision table and block-precedence rules.
- Return AUTO_APPROVE, MANUAL_REVIEW, or BLOCK with structured reasons.
- Build the approval screen prototype.

Day 8 - End-to-End Slice (Thu 22 Oct)

- Manual proposal -> evaluation -> decision -> PayPal authorization -> capture/void -> audit dashboard.
- Demonstrate happy path plus over-cap, Poor, and Insufficient data cases.
- Stop and resolve blockers before starting P1 work.

Day 9 - AG Grid Dashboard (Fri 23 Oct)

- Replace the temporary activity list with AG Grid.
- Show proposal, score, decision, exact payment status, amount, explanation, and timestamps.

Day 10 - Dashboard and Resilience (Sat 24 Oct)

- Add filters, sorting, badges, detail drawer, empty/error states, and loading behavior.
- Test duplicate payment requests and third-party failures. Begin Google Calendar OAuth only after core checks pass.

Day 11 - Calendar Proposal Prefill (Sun 25 Oct)

- Read one Google Calendar source and add Create proposal from event.
- Prefill event title/date only; the user supplies product name and proposed price. Do not infer products.

Day 12 - Bryntum Calendar (Mon 26 Oct)

- Visualize spending windows and calendar events in Bryntum.
- Link events to the proposal prefill action; preserve the manual flow as fallback.

Day 13 - KERNEL Trust Check (Tue 27 Oct)

- Implement a basic pass/fail/unavailable merchant signal with timeout behavior.
- Persist evidence/reference metadata and show the signal in evaluation details.

Day 14 - Manual Approval (Wed 28 Oct)

- Complete approve/reject UI for Fair and Insufficient data results.
- Ensure final approval gates authorization/capture and rejection is audited.

Day 15 - Integration Day (Thu 29 Oct)

- Run the complete decision matrix and exact payment-state transitions.
- Test cap boundaries, currency mismatch, duplicate requests, provider timeouts, void after authorization, and blocked capture.

Day 16 - UI and Error Polish (Fri 30 Oct)

- Polish responsive states, accessibility, errors, loading, and judge-facing explanations.
- Freeze features. Refund and all extra integrations remain out unless the core is fully stable.

Day 17 - Demo Rehearsal (Sat 31 Oct)

- Record raw footage showing one successful authorization/capture and one block/manual review.
- Verify audit records and sponsor-tool visibility. Bug fixes only.

Day 18 - Documentation (Sun 1 Nov)

- Finish README, .env.example, migration/setup steps, architecture diagram, sandbox disclosure, and API examples.
- Document Quality Score thresholds, decision table, state machine, and known limitations.

Day 19 - Deploy and Package (Mon 2 Nov)

- Test the Render backend/API deployment and chosen frontend host.
- Finalize a sub-three-minute video and Devpost text. Name only integrations actually demonstrated.

Day 20 - Core Freeze (Tue 3 Nov)

- Full team review, smoke test, backup demo recording, and submission checklist.
- Use the remaining buffer through 12 November for reliability, documentation, and submission - no speculative features.

Ownership

Owner	Primary responsibility
Dave	Scope control, coordination, demo narrative, video, and Devpost.
Henry	Next.js UI, AG Grid, Bryntum, settings, approval flow, and polish.
Celestine & Steph	Supabase schema/Auth support, constraints, PayPal, Channel3, orchestration, audit, KERNEL, and Render deployment.

Critical gates

Gate	Required result
End Day 3	Schema, Auth, constraints tests, and PayPal state contract agreed.
End Day 6	Quality Score v1 implemented with boundary and insufficient-data tests.
End Day 8	Working manual end-to-end slice. P1 work is blocked until this passes.
End Day 15	Decision matrix and payment transitions pass integration tests.
End Day 16	Feature freeze; only fixes, docs, deployment, and demo work.
3-12 Nov	Submission buffer and reliability work; final deadline is 12 Nov at 9:00 PM Lagos time.

Document version: 14 October 2026 - Locked implementation plan